Household Food Security and Structural Improvements through the Hogares Saludables Program: Evidence from Colombia

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Abstract

Housing deficits remain a global challenge with significant implications for health, well-being, and food security. In Colombia, four million households face a qualitative housing deficit, and one in four people experience food insecurity. This study evaluates the Hogares Saludables Program, which directly intervenes vulnerable households by improving their kitchens, floors, and bathrooms. As part of a series of studies that analyze the effects on a variety of household outcomes, this research focuses on whether housing improvements enhance food security. We estimate a Difference-in-Difference model using an experience-based Household Latent Food Security Index constructed using Item Response Theory, specifically a Partial Credit Model, applied to survey responses on food access, coping strategies, and related experiences. We find an improvement of 0.094 points on the 0-1 HLFSI, signaling a moderate yet meaningful development.

1 Introduction

New and rapid urbanization patterns are reshaping demography worldwide. As of 2025, an estimated 55 percent of the global population lives in urban areas. This share is expected to grow, with an additional 1.6 billion people migrating into urban areas over the next five years (Behr et al., 2021). Increased economic opportunities, higher standards of living, and labor conditions are determinants that explain the rural to urban migration (Brakarz & Jaitman, 2013).

In this context, the current housing offer is not able to meet the demand, result of the rising of migration into cities, which places greater pressure on informal settlements (Behr et al., 2021). In fact, UN-Habitat projects that up to 40 percent of the world's population will require access to adequate housing over the next five years (UN Habitat, 2025). Adequate living conditions are essential for development and well-being of societies. Poor conditions of housing and infrastructure are also related to health and nutrition outcomes, as inadequate water, sanitation, and living environments contribute to higher probability of diseases and reduced food security (Odoms-Young et al., 2024; Prüss-Ustün et al., 2019).

Colombia is a clear example of these dynamics, as a country that has experienced both a quantitative and qualitative housing deficit persistently. The quantitative dimension refers to the shortage of dwellings with structural deficiencies which make it necessary to add new stock of houses instead of improving the specific housing unit.These are cases in which families lack access to housing altogether or must share overcrowded spaces. On the other hand, the qualitative housing deficit represents dwellings that fail to meet standards of habitability, safety, or dignity. This often includes inadequate construction materials, the lack of essential services, and unsafe facilities that require structural improvements instead of new construction (DANE, 2021).

In Colombia, the quantitative deficit is estimated at 6.8% or approximately 41,000 homes as of 2024, while the qualitative deficit is higher at around 20%, affecting approximately 3.6

million households nationwide. While some families face the challenge of needing a completely new home, a much larger share lives in dwellings affected by qualitative deficiencies that directly undermine health and daily activity, especially in kitchens, access to water, and sanitation. (Torres et al., 2024). Beyond their immediate effects, these obstacles also create long term challenges, such as nutrition and food security.

The Hogares Saludables (HS) Program was developed in response to Colombia's persistent housing challenges, focusing on the qualitative deficit. Implemented in Medellín, Cali, and Barranquilla, Colombia's second, third, and fourth largest cities, the program seeks to improve the structural conditions of inadequate households by applying two complementary components: (i) reducing the qualitative housing deficit and (ii) fostering the development of work and life skills. The first component seeks to improve infrastructural deficiencies by providing materials and labor for changes to floors, kitchens, and bathrooms. The second component promotes work and life skills among beneficiaries, strengthening household capacities beyond physical upgrades.

Evidence from prior interventions show that even moderate housing upgrades can return gains in health, time use, and household welfare (Cattaneo et al., 2009; Devoto et al., 2012). Building on this evidence, we hypothesize that structural improvements may also shape household food security. Healthier living environments reduce exposure to disease, safer kitchens provide better conditions for food handling, and reduced time pressures create opportunities to generate income. These mechanisms suggest that housing improvements function as an indirect but powerful determinant of food security. By connecting the housing deficit with food security, this study highlights a non-monetary channel that is often overlooked in policy debates.

The paper is organized as follows. The next section reviews the broader literature and outlines the theoretical mechanisms linking housing improvements to food security, arguing for the importance of studying housing-related outcomes beyond the effects of household income. The third section describes the Hogares Saludables Program and presents the study

design. The fourth section introduces the empirical framework, detailing the econometric model and the data used. The section after presents the main results. Finally, the sixth section discusses the policy and research implications and concludes.

2 Housing improvement and food security

2.1 Background

In 2024, 6.8 percent of Colombian households faced a quantitative housing deficit, while 20 percent a qualitative deficit. Among the households classified under a qualitative deficit, approximately 490 thousand were affected by inadequate kitchens or unsafe cooking water (Torres et al., 2024). Previous studies on the housing deficit, such as Garcia et al. (2025), show that structural upgrades can generate multidimensional household gains, including intra-family dynamics and both mental and physical health. These dynamics are explained through subjective assessments of housing satisfaction, which stand as significant and consistent predictors of positive family outcomes.

Food security is another dimension comprised in the well-being of a household. It may be directly affected by inadequate kitchens, unsafe water, unhygienic floors or bathrooms, which are improved through the HS program. Food security is measured beyond income levels or food prices. It includes the environment where food is prepared, stored, and consumed. In this sense, Colombia provides an interesting case. With millions of households facing deficient kitchens and unsafe cooking water, it becomes essential to assess if targeted structural upgrades can translate into measurable improvements in food security.

It has been demonstrated that housing improvements enhance health, time allocation, and the home environment, which may translate into food security gains. (Cattaneo et al., 2009; Devoto et al., 2012). Likewise, evidence about the high probability of disease in a household associated with inadequate Water, Sanitation, and Hygiene (WASH) underscores how poor living conditions contribute to nutrition-related morbidity and economic difficul-

ties, restricting both food access and utilization (Prüss-Ustün et al., 2019).

Building on this rationale, we focus on structural improvements to kitchens, bathrooms, and floors in order to examine how physical modifications alone may influence food security outcomes. We adopt the Food and Agriculture Organization (FAO) definition of food security, which exists when "all people, at all times, have physical, social, and economic access to sufficient, safe, and nutritious food for an active and healthy life" (FAO, 2017).

The concept is based on four pillars: availability, access, utilization, and stability. Availability refers to the physical presence of food; for instance, reducing waste because of better storage systems leaves more edible and safer food. Access captures the ability to obtain food economically and physically. In that sense, fewer expenses related to illness or additional paid work increases household purchasing power to obtain more food. The third pillar is utilization, which relates to the safe preparation and handling of food, where cleaner surfaces and safer water reduce contamination risks. Finally, stability concerns the reliability of the previous pillars. For example, lower levels of waste and fewer health shocks can help smooth consumption patterns across months.

To put this framework into practice, we use the experience-based measurement approaches, which have been traditional in the food security field. Indicators like the Household Food Insecurity Access Scale (HFIAS) and the Food Insecurity Experience Scale (FIES) are used as alternatives to resource-based indicators of food security (Deitchler et al., 2010; FAO, 2013; Kar et al., 2018). Mainly, these instruments rely on the household's perceptions in terms of their experiences of uncertainty, compromised diets or limited availability of food. Therefore, they capture the lived reality of food insecurity more directly than availability or expenditure data on its own. Traditional implementations typically add affirmative responses into a raw score, but this method assumes that all items contribute equally and ignores differences in their severity (Cafiero et al., 2016; Deitchler et al., 2010).

Item Response Theory (IRT) models, and in particular the Partial Credit Model (PCM), improve such limitations by estimating differential item weights and mapping responses

onto a latent continuum of food insecurity (Charamba et al., 2023; Nord, 2014). In that sense, IRT/PCM models recognize that strategies such as skipping meals or selling assets are signs of more severe food insecurity than relying on credit. Then, it positions households along a continuous scale that reflects both the likelihood and intensity of their experiences. By adopting this framework, this study constructs a psychometrically grounded index that provides greater comparability across households and avoids the arbitrary equal weighting.

2.2 Mechanisms

We identify three primary mechanisms through which kitchen, bathroom, and floor upgrades may influence influence the latent trait of household food security: (i) sanitation and hygiene, where cleaner surfaces and improved water access reduce food-borne risks and nutrition-related illness; (ii) psychosocial state, where a safer, cleaner, and more organized cooking environment alleviates stress; and (iii) food handling and storage, where better storage conditions mitigate losses from pests and other forms of waste.

The upgrades to floors, bathrooms, and kitchens are expected to improve household hygiene and food safety. By creating cleaner environments for food preparation and sanitation, households have reduced exposure to pathogens that cause food-borne and enteric illnesses, such as diarrhea (Esrey et al., 1991; Curtis & Cairncross, 2003; Fewtrell et al., 2005; Thomson et al., 2009; Brakarz & Jaitman, 2013). Therefore, the reduction of food-borne illness decreases medical expenses and limits health shocks, (Gertler & Gruber, 2002; Xu et al., 2003)strengthening the pillars of access and stability.

The second mechanism is explained through psychosocial bandwidth, which refers to the mental health effects that influence household food security. The HS upgrades can reduce stress and enhance overall well-being (Cattaneo et al., 2009; Thomson et al., 2009). Improved mental health is shown to increase cognitive capacity, allowing households to reason more effectively and make better and planned decisions. Specifically, greater bandwidth is expected to permit more planning and budgeting for household expenses, including food

and nutrition (Shah et al., 2015; Mani et al., 2013; Ridley et al., 2020). By strengthening these decision-making processes, households enhance their access, stabilize consumption, and optimize the utilization of resources, resulting in the improvement food security.

The third mechanism operates through food storage, primarily affected by kitchen upgrades. Improved kitchen infrastructure facilitates more appropriate refrigeration and storage of perishable foods such as meat and vegetables, reducing contamination risks and limiting pest presence in the household (Bingham & Hagstrum, 2023; Guru et al., 2022). Furthermore, enhanced storage and refrigeration reduce food waste by preserving food for longer periods (Bajželj et al., 2020; Martindale & Schiebel, 2017).

When taken together, these mechanisms clearly paint the picture on how structural housing upgrades extend beyond physical improvements to affect the lived experiences of food security. By reducing exposure to illness, reducing stress, and improving food handling and storage, the Hogares Saludables Program is expected to move households toward greater food security along this latent continuum, highlighting the potential of housing interventions to generate broader welfare gains.

3 Program and Study Design

3.1 The Hogares Saludables Program

The Hogares Saludables program, led by Cementos Argos, aims to reduce the qualitative housing deficit among vulnerable urban households in Colombia. Its main objective is to improve health, well-being, and social outcomes by addressing poor living conditions, while equipping families with practical skills that strengthen their livelihood. The program provides structural upgrades to household areas that are linked to health and daily functioning, more specifically by replacing dirt floors, unsafe cooking spaces, and inadequate sanitation facilities with hygienic alternatives.

3.2 Evaluation and Assignment Strategy

From its initial planning, the program's ambition to reach families facing housing vulnerabilities had significant methodological and ethical restrictions. In the original evaluation design, the ideal scenario wanted to create a two-stage randomized experiment: first, assigning neighborhoods to treatment and control groups through cluster-based randomization, and second, selecting households randomly within those clusters. This would have maximized causal identification while minimizing potential contamination between treated and untreated areas.

However, logistical constraints, ethical considerations, and the program's community-based implementation strategy did not make this design practical. Therefore, Hogares Saludables employed a quasi-experimental design based on a Difference-in-Differences (DiD) framework combined with matched comparison areas. Using data from the 2018 National Population and Housing Census, the research team conducted a cluster analysis that grouped census blocks along four broad dimensions of vulnerability: housing conditions, demographic composition, educational attainment, and labor market indicators. This procedure allowed to identify neighborhoods with comparable social and structural profiles, mainly focusing on those with the highest qualitative housing deficits.

Within the most vulnerable cluster, the research team selected neighborhoods in Medellín, Cali, and Barranquilla. Then, the team had field visits to verify eligibility, making sure that each selected household met the criteria related to structural integrity, access to public services, and tenure security. Approximately six hundred households were ultimately assigned to each group, producing a balanced sample suitable for rigorous statistical comparison.

The final design merged an ethical and rigorous methodology. By integrating spatial matching, field verification, and longitudinal data collection, the evaluation established a robust framework for estimating the program's causal effects. This approach enabled the identification of meaningful improvements in housing quality and well-being while upholding

the ethical imperative to prioritize families in greatest need.

3.3 Survey Design

Households answered a survey which consisted of 235 questions across seven modules: personal characteristics, housing, food security, income, community environment, mental health, and family environment. To mitigate potential confounding, several regressions include controls for household composition and demographics: the number of residents, the proportion of female members, average age, the share of members with fewer than six years of education, the proportion of adults over 65, and the unemployment rate among household members. Regressions also include a dummy variable for participation in the Hogares Saludables program to account for sampling differences and to test for potential anticipation effects, which were never statistically significant.

We considered directly asking households to report income as unreliable, considering that earnings often fluctuate monthly and gang presence discourages disclosure. To address this, the survey adopted a more neutral approach, asking households to place themselves within one of seven consumption expenditure brackets, which are used as controls in the regressions. In addition, regressions control for whether the household is classified as living in moderate or extreme poverty according to the Colombian government's Sistema de Identificación y Clasificación de Potenciales Beneficiarios para Programas Sociales (SISBEN) database.

Geographic Information System (GIS) software was used to calculate the maximum slope surrounding each dwelling, measured in degrees across eight cardinal directions at 50-meter intervals. The analysis also computed the distance from each dwelling to the central business district (CBD) and to the nearest transit station (funicular, metrocable, subway, or bus), hospital, or police station.

4 Empirical Framework

4.1 Econometric modeling

Given that we observe unbalanceness in critical characteristics, the evaluation of the Hogares Saludables program adopts a quasi-experimental design. The main quantitative identification strategy is a Differences-in-Differences estimator combined with propensity score matching (PSM). This approach compares changes in outcomes for beneficiary households (treatment group) with changes in outcomes for a comparable set of households not exposed to the program (control group), between baseline and follow-up measurements. Under the assumption of parallel trends, the interaction term between treatment status and post-intervention period captures the causal impact of the intervention (Angrist & Pischke, 2008). Absent the HS program, treatment group would be in a similar food security situation as the control group households.

4.1.1 Identification Strategy

We estimate a weighted two-period Difference-in-Differences model with section fixed effects and an explicit treatment main effect. The econometric specification is given by:

$$Y_{it} = \alpha + \tau \text{Treat}_i + \gamma \text{Post}_t + \delta(\text{Treat}_i \times \text{Post}_t) + \mu_{s(i)} + \varepsilon_{it}, \tag{1}$$

where Y_{it} denotes the outcome for household i in period t; Treat_i is a binary indicator equal to one for households that received the Hogares Saludables (HS) intervention; Post_t is a post-intervention dummy equal to one for observations from the endline survey; and $\text{Treat}_i \times \text{Post}_t$ is the interaction term capturing the DiD estimator. The term $\mu_{s(i)}$ represents section (cluster) fixed effects, which absorb unobserved, time-invariant heterogeneity at the level of the randomization unit.

The model is estimated by weighted least squares, where w_i are the propensity-score

matching weights:

$$\min_{\alpha,\tau,\gamma,\delta,\{\mu_s\}} \sum_{i,t} w_i \left(Y_{it} - \alpha - \tau \text{Treat}_i - \gamma \text{Post}_t - \delta(\text{Treat}_i \times \text{Post}_t) - \mu_{s(i)}\right)^2. \tag{2}$$

The coefficient of interest is δ, which provides the DiD estimate of the program's effect on household outcomes under the parallel-trends assumption conditional on section fixed effects. In other words, in the absence of the Hogares Saludables intervention, treated and control households would have experienced similar average changes in Y between baseline and endline, once conditioned on section fixed effects and propensity-score matching weights. The weighting procedure improves balance in observable characteristics, while the fixed effects account for time-invariant unobservables at the cluster level.

This specification includes section fixed effects ($\mu_{s(i)}$) but does not incorporate household fixed effects. Consequently, the treatment indicator Treat_i is identifiable and appears explicitly in the model. The inclusion of section fixed effects controls for unobserved heterogeneity at the level of the intervention's assignment unit, mitigating potential bias from contextual differences across neighborhoods.

4.1.2 Propensity Score Matching

Let $Y_{i,t}$ be an outcome variable for household i at time t, conditional on treatment assignment $T_i \in \{0, 1\}$. The average treatment effect on the treated (ATT) is estimated as:

$$E(\Delta Y^1_{i,t+1} - \Delta Y^0_{i,t+1} \mid T_i = 1, P(Z_i)) = E(\Delta Y^1_{i,t+1} \mid T_i = 1, P(Z_i)) - E(\Delta Y^0_{i,t+1} \mid T_i = 1, P(Z_i)),$$

where Δ denotes the before–after change in outcome and $P(Z_i)$ is the estimated probability of being treated conditional on a vector of baseline characteristics Z_i.

The propensity score is estimated from a logit model:

$$\Pr(T_i = 1) = F(Z_i),$$

where Z_i includes socio-economic variables such as household size, age of the household head, the share of women, elderly, adults, children, and Black household members. We also control for the share of members with primary education or less, the share with completed secondary education, an unemployment dummy, monthly household expenses, and indicators of neighborhood safety. In addition, we control for pre-treatment levels of mental and physical health, family violence, and family relations, as well as the estimated total cost of the housing intervention. Housing quality variables are excluded since they are directly affected by the treatment.

Kernel matching is applied to construct the counterfactual as a weighted average of all control households. We enforce common support, allow matching with replacement, and bootstrap standard errors with 1,000 replications following. The estimations further account for the unit of randomization and include section fixed effects.

After matching, the DiD analysis is estimated on matched households as:

$$Y_{it} = \beta_0 + \beta_1 T_i + \beta_2 Post_t + \beta_3 (T_i \cdot Post_t) + \mu_{s(i)} + \varepsilon_{it}, \tag{3}$$

where T_i is an indicator for treatment assignment, $Post_t$ is an indicator for the endline survey, and β_3 identifies the ATT.

4.1.3 Outcome Construction: Food Security Index

Household food security is measured through experience-based survey items that capture both constraints and coping behaviors: (i) perceived sufficiency of income to cover food purchases during the last month; (ii) access to safe drinking water for meal preparation; (iii) availability of covered food storage to protect against pests; and (iv) nine coping strategies employed when food or money was insufficient. These strategies include buying on credit, borrowing money, purchasing food daily, bartering, using savings, sending household members to community kitchens or school cafeterias, reducing health or education spending, selling household assets, and extending breastfeeding due to lack of food.

The income sufficiency question was treated as an ordered categorical variable, with higher values indicating greater perceived ability to afford food. Water access, covered storage, and each coping strategy were coded as binary items. All variables were recoded so that higher numeric values consistently reflect better food security—for example, answering 'no' to a negative coping strategy contributes a higher score, favorable water or storage conditions are coded higher, and the income scale is inverted so larger numbers indicate greater sufficiency.

We rely on an Item Response Theory specification to construct our main outcome variable, the Household Latent Food Security Index (HLFSI). Specifically, we use the one-factor Partial Credit Model by estimating item severity (with more severe experiences positioned at higher thresholds) and discrimination (the extent to which items differentiate households along the latent continuum), avoiding reliance on arbitrary weights. This allows us to recover a latent measure of household food security. By a latent measure, we refer to an unobservable attribute inferred from the household's responses of the experience-based survey items.

Let Y_{ij} denote the response of household i to item j. For binary items, $Y_{ij} \in \{0, 1\}$, while for ordered categorical items, Y_{ij} takes values $0, \ldots, m_j$. The PCM specifies the probability that household i selects response category k of item j as:

$$\Pr(Y_{ij} = k \mid \theta_i) = \frac{\exp\left(\sum_{s=0}^{k}(\theta_i - \beta_{js})\right)}{\sum_{t=0}^{m_j} \exp\left(\sum_{s=0}^{t}(\theta_i - \beta_{js})\right)}, \tag{4}$$

where θ_i represents the latent food security of household i, β_{js} are the estimated step thresholds for item j, and m_j is the number of response categories.

Household factor scores are extracted using expected a posteriori (EAP) estimation. These scores define the HLFSI as a standardized latent trait (θ, mean 0 and standard deviation 1), along with a rescaled version between 0 and 1 for interpretation. Higher values correspond to greater food security.

Unlike additive indices that assume equal severity across items, the PCM incorporates differences in item thresholds (e.g., selling assets versus buying on credit), ensuring that the

index reflects both the intensity and likelihood of experiences. This approach aligns with advances in food insecurity measurement (Deitchler et al., 2010; FAO, 2013; Cafiero et al., 2016; Nord, 2014).

Finally, this Household Latent Food Security Index is used as the dependent variable in the matched DiD specification described above, allowing us to estimate the causal effect of the housing intervention on latent household food security.

4.2 Data and sampling

The research draws on samples from informal settlements in Medellín, Cali, and Barranquilla, cities with populations of 2.6, 2.9, and 1.2 million, respectively. On average, 15.3 percent of households in these urban centers face severe housing deficits (DANE, 2023). Informal settlements—locally referred to as invasiones—emerged as a prominent feature in Colombia when populations displaced by armed conflict relocated to high-risk areas and constructed precarious housing. These are usually located in high-risk terrain, unsuitable for formal construction(Ocampo, 2003).

The study sample was collected in anticipation of a future evaluation of the Hogares Saludables program and comprises 1,033 homeowner households, with roughly one-third located in each of the three cities. Under the program, half of these households would eventually receive free interior home improvements, provided by Grupo Argos, the largest cement and ready-mix concrete producer in Colombia. The survey data, however, were collected prior to the intervention, and participating households were unaware of the nature or extent of the future investments.

Because participation was determined by program eligibility based on vulnerability, the evaluation estimates effects for the population actually treated. Econometrically, the DiD estimator combined with propensity score matching identifies the Average Treatment Effect on the Treated (ATT). However, given that assignment was policy-driven and not random, results should be interpreted in practice as Intention-to-Treat (ITT) associations for the

eligible population, rather than generalizable average treatment effects across all households.

The second subsample was drawn from other neighborhoods using a stratified, multi-stage sampling approach based on the 2018 National Census. Colombia's census geography is structured into three levels: sector, section, and block. Sectors served as the primary sampling unit to ensure homogeneity, after which a k-median clustering process aggregated sections with the highest levels of housing deficiencies and demographic characteristics similar to those of the treated group. Eight such clusters were then randomly selected. As in the HS sample, community organizations identified households with substandard dwellings, and participation was encouraged with vouchers worth 100,000 Colombian pesos redeemable at local stores and supermarkets. This incentive proved effective, yielding a 98 percent participation rate.

4.3 Critical stylized facts

The surveyed households illustrated the situation of urban vulnerability in Colombia, especially in families of lower socioeconomic strata. This is seen through the low levels of education achieved by household heads: slightly more than half completed only primary school, while fewer 20% graduated from secondary education. Unemployment reaches 11.3%, and 62.3% are classified as poor under SISBEN, Colombia's official poverty identification system. Additionally, the average reported spending level of 3.1 (out of 8) further shows the restricted consumption of the households, whereas perceptions of neighborhood insecurity, represented by a mean 2.4 (out of 5), underscore the social and environmental stressors inherent to Colombia's vulnerable urban areas.

On average, households are composed of three to four members, with children making up 30% of the household. 68.5% of households are headed by women, reflecting that women frequently shoulder both economic and caregiving responsibilities (Chant, 2016). Approximately 11.8% of households include at least one older adult, and 18% identify as Afro-descendant. Even though the mean age of household heads is 54 years, the variation

shows that many different ages coexist in the households.

Health-related indicators illustrate baseline conditions that these households face, which may be directly related to Food Security. While severe illness is not prevalent, the occurrence of common symptoms such as fever, diarrhea, and skin irritation shows the exposure to health risks likely caused by the inadequate conditions of the dwellings. Despite these challenges, the reported satisfaction level is relatively high, with a mean 3.88 on a 1–5 scale.

The average Household Latent Food Security Index value of 0.686 indicates that although many families meet their basic nutritional needs, a significant proportion remain vulnerable to food insecurity. The latent food security score exhibits substantial variation, exposing inequality within the sample: some households maintain relative stability, whereas others experience persistent deprivation.

Table 1: Descriptive statistics (baseline, household level)

Variable	N	Mean	SD	Min	Max
Socioeconomic variables					
Unemployed (household head)	1012	0.113	0.316	0	1
Primary education (household head)	1163	0.508	0.330	0	1
Secondary education (household head)	1163	0.191	0.251	0	1
Spending level (categorical)	1163	3.103	1.762	1	8
SISBEN poverty classification	1163	0.623	0.411	0	1
Perceived neighborhood insecurity	1163	2.404	0.908	1	5
Demographic variables					
Older adult in household	1163	0.118	0.253	0	1.000
Afro-descendant household	1163	0.179	0.361	0	1.000
Household head age	1163	54.468	15.552	0	95.000
Household size	1163	3.391	1.813	1	13.000
Children (share of household members)	1163	0.312	0.246	0	0.875
Female household head	1163	0.685	0.465	0	1.000
Outcome variables					
Share of members with diarrhea	1163	0.042	0.141	0.00	1.000
Share of members with fever	1163	0.073	0.185	0.00	1.000
Food Security Index (FSI)	1163	0.686	0.220	0.00	1.000
Latent food security score (θ)	1163	0.000	1.000	-3.12	1.429
Share of members with skin irritation	1163	0.033	0.125	0.00	1.000
Household satisfaction (1–5 scale)	968	3.883	0.792	1.00	5.000
Share of members who visited a doctor	1163	0.131	0.251	0.00	1.000
Share of members with vomiting	1163	0.034	0.127	0.00	1.000

5 Results

Table 2 reports the estimated effects of the HS program on household food security. The table presents Difference-in-Differences estimates of the program's impact on the Household Latent Food Security Index, expressed as a 0–1 rescaled index in column 1 and as a standardized latent score in column 2. All specifications incorporate section fixed effects and propensity-score matching weights, consistent with the empirical framework outlined earlier.

The time trend captured by the Post coefficient is first examined. In both outcome specifications, the Post coefficient is negative and statistically significant, suggesting that in the absence of the program, households experienced a decline in food security between baseline and endline. This finding is consistent with adverse aggregate shocks that may have reduced food access and limited the frequency of positive food-security behaviors.

Turning to the parameter of interest, the DiD interaction term is positive and consistent across both specifications. These estimates indicate that the HS intervention mitigated a substantial portion of the deterioration observed among control households and generated net improvements for treated households relative to their matched counterparts. On the interpretable 0–1 scale, the estimated effect is +0.094 points, corresponding to a 9.4-percentage-point increase along the normalized HLFSI range. When expressed in standardized units, the effect equals +0.43 standard deviations, signifying a moderate yet meaningful improvement in household food security.

Table 2: Impact on Household Food Security

	Food Security	
	HLFSI	HLFSI Theta
	(1)	(2)
Tratamiento	−0.464	−2.123
	(0.366)	(1.675)
Post	−0.111***	−0.198***
	(0.013)	(0.060)
Tratamiento × Post	0.094***	0.428***
	(0.018)	(0.081)
Observations	1,920	1,920
R^2	0.617	0.610
Adjusted R^2	0.233	0.219
Residual Std. Error	0.192	0.879
F Statistic	1.607***	1.560***

Note: *p<0.1; **p<0.05; ***p<0.01. Standard errors in parentheses.

6 Discussion and conclusion

This study examined whether structural housing improvements, specifically upgrades to kitchens, bathrooms, and floors, can enhance household food security among vulnerable urban populations in Colombia. By integrating propensity-score matching with a Difference-in-Differences approach and constructing a psychometrically grounded Household Latent Food Security Index based on Item Response Theory, the analysis identifies a plausible causal

pathway through which non-monetary interventions influence nutrition-related outcomes.

The results indicate that the Hogares Saludables program did have a significant improvement in food security outcomes. The estimated increase of 0.094 points in the normalized HLFSI, suggests that physical housing upgrades had an effect in the experience of food security in the treated households, which the control households during the same period did not experience. This is consistent with evidence that shows that housing improvements return health and well-being benefits extending beyond material comfort (Cattaneo et al., 2009; Devoto et al., 2012; Thomson et al., 2009). In the Colombian context, where the qualitative housing deficit is prevalent, these findings are particularly important for urban social policy.

The negative time trend in food security, captured by the Post coefficient, indicates that in the absence of the intervention, participating households would have experienced reductions in food access and utilization. The HS program's ability to counteract these dynamics underscores the resilience potential of housing interventions, even when implemented on a relatively modest scale.

Three mechanisms may account for these effects. First, improvements in sanitation and hygiene reduce exposure to pathogens and food-borne illnesses, indirectly protecting household budgets from health shocks and enabling more stable food access. Second, the psychosocial relief associated with safer, cleaner, and more organized kitchens may alleviate stress and cognitive burden, enhancing decision-making regarding expenditure, nutrition, and time allocation. Third, improved storage and preparation facilities reduce food spoilage and waste, thereby strengthening the reliability of food availability within the household. Collectively, these mechanisms underscore that housing is not only a determinant of physical health but also a fundamental component of food security.

Several limitations are to be considered. The analysis focuses on a specific subset of vulnerable urban homeowners, which limits the generality of the findings. Additionally, the short follow-up period restricts our capacity to assess the long-term sustainability of improvements in food security. Future research should include additional data waves and

explore complementary pathways such as time use, health expenditures, and dietary diversity to capture the multidimensional benefits of improved housing quality.

From a policy perspective in Colombia, where policy frameworks frequently treat housing and nutrition as separate domains, the Hogares Saludables model illustrates the potential of cross-sectoral interventions that utilize physical infrastructure as a platform for human development. By linking the right to adequate housing with the right to adequate food, the program provides a scalable framework for multidimensional poverty reduction.

In conclusion, structural housing improvements constitute more than comfort-oriented investments. They are essential to human capability and resilience. Hogares Saludables enhances not only living conditions but also household food security through mechanisms related to hygiene, psychological well-being, and resource optimization. As urbanization accelerates and inequality intensifies, acknowledging housing as a determinant of nutritional well-being becomes crucial for designing holistic, sustainable, and equitable social policy.

7 Use of Generative Artificial Intelligence

Generative Artificial Intelligence tools, specifically ChatGPT, was used for purposes of improving drafting aspects such as grammar, coherence, and flow.

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